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SIMATS ENGINEERING

Feature Detection and Matching - Level 2

Score: 25 / 25

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Answer Details

Q1: For the Harris matrix M=[[100,20],[20,50]], what is the trace of M?

A. 50

B. 70

C. 100

D. 150

Submitted Answer: 150

Correct Answer: 150

Q2: An image contains 1000 pixels, and each pixel is represented using 8 bits. How much raw storage is required?

A. 800 bits

B. 8000 bits

C. 80000 bits

D. 1000 bits

Submitted Answer: 8000 bits

Correct Answer: 8000 bits

Q3: What is the primary purpose of image preprocessing in computer vision?

A. To increase the image file size

B. To improve the image quality and prepare it for further processing

C. To remove all image information

D. To convert every image into a binary image

Submitted Answer: To improve the image quality and prepare it for further processing

Correct Answer: To improve the image quality and prepare it for further processing

Q4: In Harris Corner Detection, what does the structure matrix represent?

A. Color information of the image

B. Local intensity variations in different directions

C. Image file size

D. Number of objects in the image

Submitted Answer: Local intensity variations in different directions

Correct Answer: Local intensity variations in different directions

Q5: Which of the following is an example of image enhancement?

A. Histogram equalization

B. Object classification

C. Feature matching

D. Image segmentation

Submitted Answer: Histogram equalization

Correct Answer: Histogram equalization

Q6: A Harris matrix has one large eigenvalue and one eigenvalue close to zero. What type of structure does this represent?

A. Corner

B. Edge

C. Flat region

D. Noise-free background

Submitted Answer: Edge

Correct Answer: Edge

Q7: What is the primary purpose of PCA in image feature extraction?

A. Increase the dimensionality of image data

B. Reduce dimensionality while retaining important information

C. Detect straight lines

D. Detect image edges directly

Submitted Answer: Reduce dimensionality while retaining important information

Correct Answer: Reduce dimensionality while retaining important information

Q8: What is the main objective of Harris Corner Detection?

A. Detect regions with strong intensity changes in multiple directions

B. Compress an image into JPEG format

C. Remove image noise only

D. Convert RGB images into grayscale images

Submitted Answer: Detect regions with strong intensity changes in multiple directions

Correct Answer: Detect regions with strong intensity changes in multiple directions

Q9: What is the main purpose of edge detection?

A. To identify boundaries where intensity changes significantly

B. To increase image resolution

C. To remove all corners

D. To convert an image into a text document

Submitted Answer: To identify boundaries where intensity changes significantly

Correct Answer: To identify boundaries where intensity changes significantly

Q10: What is the major advantage of SIFT features?

A. They are invariant to scale and rotation

B. They work only with binary images

C. They eliminate the need for feature descriptors

D. They can detect only horizontal edges

Submitted Answer: They are invariant to scale and rotation

Correct Answer: They are invariant to scale and rotation

Q11: A Harris detector produces R1=2500, R2=-1200, and R3=0.5. Which classification is most appropriate?

A. R1 Corner, R2 Edge, R3 Flat Region

B. R1 Edge, R2 Corner, R3 Flat Region

C. R1 Flat Region, R2 Edge, R3 Corner

D. All three are corners

Submitted Answer: R1 Corner, R2 Edge, R3 Flat Region

Correct Answer: R1 Corner, R2 Edge, R3 Flat Region

Q12: An image contains significant intensity changes in the horizontal direction but very little change in the vertical direction. How is the region most likely classified?

A. Corner

B. Edge

C. Flat region

D. Circular feature

Submitted Answer: Edge

Correct Answer: Edge

Q13: Consider the Harris matrix M=[[100,20],[20,50]]. What is det(M)?

A. 4600

B. 5000

C. 5400

D. 6000

Submitted Answer: 4600

Correct Answer: 4600

Q14: Which image representation stores each pixel using a single intensity value?

A. RGB image

B. Grayscale image

C. Multispectral image

D. Vector image

Submitted Answer: Grayscale image

Correct Answer: Grayscale image

Q15: A corner in an image is generally characterized by what?

A. Little or no intensity variation in all directions

B. Significant intensity variation in only one direction

C. Significant intensity variation in multiple directions

D. Constant pixel intensity throughout the image

Submitted Answer: Significant intensity variation in multiple directions

Correct Answer: Significant intensity variation in multiple directions

Q16: A Harris matrix has two large positive eigenvalues. What does this indicate?

A. Flat region

B. Edge

C. Corner

D. Uniform background

Submitted Answer: Corner

Correct Answer: Corner

Q17: The same object appears in two images, but its size and orientation have changed. Which technique is appropriate for finding corresponding keypoints?

A. Sobel Edge Detection

B. SIFT

C. Mean Filtering

D. Thresholding

Submitted Answer: SIFT

Correct Answer: SIFT

Q18: Which operator is commonly used for edge detection?

A. Sobel operator

B. PCA

C. SIFT descriptor

D. Hough accumulator

Submitted Answer: Sobel operator

Correct Answer: Sobel operator

Q19: Which technique is particularly suitable for detecting straight lines and circles?

A. PCA

B. Hough Transform

C. Histogram Equalization

D. Gaussian Blur

Submitted Answer: Hough Transform

Correct Answer: Hough Transform

Q20: Which of the following is the Harris corner response equation?

A. R=det(M)-k(trace(M))^2

B. R=det(M)+k(trace(M))^2

C. R=trace(M)-kdet(M)

D. R=det(M)*trace(M)

Submitted Answer: R=det(M)-k(trace(M))^2

Correct Answer: R=det(M)-k(trace(M))^2

Q21: A line in an image is represented by several edge points. Which technique can accumulate votes to determine the parameters of the line?

A. PCA

B. SIFT

C. Hough Transform

D. Histogram Equalization

Submitted Answer: Hough Transform

Correct Answer: Hough Transform

Q22: What is the primary purpose of feature extraction?

A. To represent important and distinctive information from an image

B. To increase the number of pixels

C. To delete all image boundaries

D. To convert every image into a binary image

Submitted Answer: To represent important and distinctive information from an image

Correct Answer: To represent important and distinctive information from an image

Q23: Using k=0.04, calculate the Harris response for M=[[100,20],[20,50]].

A. 4400

B. 3700

C. 4600

D. 4800

Submitted Answer: 3700

Correct Answer: 3700

Q24: An image feature vector has 100 dimensions, but most information is concentrated in fewer directions. Which technique can reduce dimensionality while preserving maximum variance?

A. Hough Transform

B. Harris Corner Detection

C. PCA

D. Sobel Filtering

Submitted Answer: PCA

Correct Answer: PCA

Q25: A 3x3 image neighborhood has horizontal and vertical gradients Ix=4 and Iy=3. What are Ix^2, Iy^2, and IxIy?

A. 16, 9, 12

B. 8, 6, 7

C. 12, 16, 9

D. 7, 12, 16

Submitted Answer: 16, 9, 12

Correct Answer: 16, 9, 12